Noman Bin Morshed

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Education

University of Dhaka, Bangladesh **2022–Present**
Bachelor of Science (Honours) in Nuclear Engineering
Key Coursework: Computer Programming (C++), Nuclear Physics, Statistics, Numerical Methods, Nuclear Instrumentation, MATLAB Modeling, Nanoscience, Quantum Mechanics, Nuclear Fusion.
CGPA: 3.73/4.00

Experience

IBM Quantum – Qiskit Advocate ***Remote***

- Completed the official Qiskit Advocate Training Program.
- Earned multiple IBM Quantum Learning badges covering quantum algorithms, circuits, and noise models.
- Contributed to community-led research on quantum simulation and circuit optimization.

Bangladesh Atomic Energy Commission (BAEC) ***Industrial Training***

- Gained hands-on experience in nuclear material characterization and nondestructive evaluation (NDE) techniques.
- Performed and analyzed:
  - **Eddy Current Testing (ECT)** – for surface and near-surface defect detection.
  - **Magnetic Particle Testing (MPT)** – for identifying flaws in ferromagnetic materials.
  - **Radiographic Testing (RT)** – for internal discontinuity inspection using ionizing radiation.
  - **Ultrasonic Testing (UT)** – for subsurface flaw detection and material property estimation.

Research Projects

Quantum Galton Board Simulator

- Developed quantum circuits for normal, exponential, and Hadamard walk distributions based on the Carney-Varcoe Universal Statistical Simulator framework using `Qiskit`.
- Engineered diffusion-mimicking circuit logic employing Pauli-X initialization, ancilla-driven amplitude control, and localized spatial mapping via Controlled-SWAP (CSWAP) gate routing.
- Evaluated execution viability under realistic hardware noise configurations by benchmarking distributions against the `FakeGeneva` IBM backend noise model.
- Mitigated fidelity degradation in scaling architectures (maintaining up to 90% fidelity under 10 qubits) by deploying standard and AI-assisted circuit transpilation optimization paths.

Publications

Simulated Quantum Annealing-Genetic ALgorithm Framework for Nuclear Fuel Loading Optimization

- Developed a three-layer hybrid framework integrating quantum-inspired metaheuristics, deep-learning surrogate modeling, and evolutionary search to solve the NP-hard pressurized water reactor (PWR) core fuel loading pattern optimization problem.
- Cast the multi-constraint fuel reloading placement as a 579-variable Quadratic Unconstrained Binary Optimization (QUBO) problem using `PyQUBO`, leveraging Simulated Quantum Annealing (SQA) to generate diverse, constraint-satisfying seed populations.
- Trained a multi-output residual Convolutional Neural Network (CNN) surrogate with positional encoding and squeeze-and-excitation attention on 10,000 Monte Carlo simulations using `OpenMC` to predict critical parameters (k_{eff}, F_a, F_q) simultaneously.
- Implemented a count-preserving Genetic Algorithm (GA) accelerated by the CNN surrogate for batched fitness evaluation, reducing physics evaluation times from 12 hours to under a second while achieving a reduction in peak power factors.

Quantum-Mechanical Study of the Electronic Properties of $U_xPu_yO_z$ Compounds

- Conducted first-principles DFT simulations using Quantum ESPRESSO for uranium-based mixed oxides.
- Calculated and benchmarked total energy, band gap, DOS/PDOS, stress, and force parameters.
- Investigated structure–property relationships to identify materials with optimal mechanical and electronic performance.
- Compared simulation results with literature data for validation and benchmarking.

Technical Skills

Quantum & Simulation:	DWAVE, OpenJij, OpenMC, Pennylane, Qiskit, Quantum ESPRESSO,
Programming & Tools:	Python, MATLAB, C++, Git, Linux, Jupyter Notebook
Engineering & Modeling:	ANSYS HFSS, KiCad, PyGAD, Tensorflow

Additional Coursework

2025 Quantum Program

- Offered by: The Washington Institute for STEM, Entrepreneurship and Research
- Focus Areas: Functional Quantum Linear Solvers, Solving Partial Differential Equations, Quantum Simulation and Lie Theory, Quantum Algorithms for Nonlinear Differential Equations

MTV Nuclear Engineering Summer School 2025

- Offered by: University of Michigan
- Focus Areas: Radiation Detection, Radiation Imaging, Scintillator Synthesis, Monte Carlo Simulation Methods